Problem 2 (Characterization via level-1 Fourier weight). Let $f : \{-1,1\}^n \to \{-1,1\}$ be a Boolean function. Define the level-1 Fourier weight

$$W^1[f] = \sum_{i=1}^{n} \widehat{f}(\{i\})^2.$$

Prove that if

$$W^1[f] = 1,$$

then f is a dictator function $f(x) = x_i$ or its negation $f(x) = -x_i$, for some $i \in [n]$.

Thought for 54s ›

Let $\chi_S(x) = \prod_{j \in S} x_j$ be the Fourier characters on $\{-1,1\}^n$. Then

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S)\chi_S(x),$$

and Parseval says

$$\sum_{S \subseteq [n]} \hat{f}(S)^2 = \mathbb{E}[f(x)^2] = 1,$$

since $f(x) \in \{-1,1\}$.

You are given that

$$W^1[f] = \sum_{i=1}^{n} \hat{f}(\{i\})^2 = 1.$$

So the entire Fourier mass is already on level 1. Therefore every other Fourier coefficient must be 0:
$\hat{f}(\emptyset) = 0$, and $\hat{f}(S) = 0$ for all $|S| \geq 2$.

Hence f has the form

$$f(x) = \sum_{i=1}^{n} a_i x_i, \qquad \text{where } a_i := \hat{f}(\{i\}),$$

with

$$\sum_{i=1}^{n} a_i^2 = 1.$$

Now use the fact that f is Boolean, so $f(x)^2 = 1$ for every x. Squaring the linear form gives

$$1 = f(x)^2 = \left(\sum_{i=1}^{n} a_i x_i\right)^2 = \sum_{i=1}^{n} a_i^2 + 2\sum_{i<j} a_i a_j x_i x_j.$$

Since $\sum_i a_i^2 = 1$, this becomes

$$1 = 1 + 2\sum_{i<j} a_i a_j x_i x_j,$$

so

$$\sum_{i<j} a_i a_j x_i x_j = 0 \qquad \text{for every } x \in \{-1,1\}^n.$$

But the functions $x_i x_j = \chi_{\{i,j\}}(x)$ are distinct Fourier characters, hence linearly independent. Therefore every coefficient must vanish:

$$a_i a_j = 0 \qquad \text{for all } i \neq j.$$

So at most one a_i is nonzero. Since $\sum_i a_i^2 = 1$, exactly one is nonzero, say a_k, and then $a_k^2 = 1$, so $a_k = \pm 1$.

Thus

$$f(x) = a_k x_k = x_k \quad \text{or} \quad f(x) = -x_k.$$

So f is a dictator or anti-dictator.

Problem 3 (Properties of recursive majority). Let n be an **odd** integer. Define the *recursive majority function of depth d*

$$f : \{-1,1\}^{n^d} \to \{-1,1\}$$

as follows: Partition the input into n blocks, compute the recursive majority on each block of depth $d-1$, and output the majority of these values.

Determine whether f satisfies each of the following properties:

1. Monotone
2. Odd (i.e., $f(-x) = -f(x)$)
3. Unanimous (i.e., $f(1,1,\ldots,1) = 1$ and $f(-1,-1,\ldots,-1) = -1$)
4. Symmetric

Thought for 2m 18s ›

Let RM_d denote the recursive majority of depth d on n^d bits, where n is odd. Write the input as $x = (x^{(1)}, \ldots, x^{(n)})$, with each block $x^{(j)} \in \{-1,1\}^{n^{d-1}}$. Then

$$RM_d(x) = \mathrm{Maj}_n\left(RM_{d-1}(x^{(1)}), \ldots, RM_{d-1}(x^{(n)})\right),$$

with $RM_1 = \mathrm{Maj}_n$.

Let $m = \frac{n+1}{2}$, so "majority" means at least m many $+1$'s.

1. Monotone: Yes

Majority on an odd number of bits is monotone: if $u_i \leq v_i$ for all i, then $\mathrm{Maj}_n(u) \leq \mathrm{Maj}_n(v)$.

Now argue by induction on d.

- Base case $d = 1$: $RM_1 = \mathrm{Maj}_n$, so it is monotone.
- Inductive step: assume RM_{d-1} is monotone. If $x \leq y$ coordinatewise, then for each block,

  $$RM_{d-1}(x^{(j)}) \leq RM_{d-1}(y^{(j)}).$$

  Applying monotonicity of Maj_n,

  $$RM_d(x) = \mathrm{Maj}_n(RM_{d-1}(x^{(1)}), \ldots, RM_{d-1}(x^{(n)})) \leq \mathrm{Maj}_n(RM_{d-1}(y^{(1)}), \ldots, RM$$

So RM_d is monotone for every d.

2. Odd: Yes

Again use induction. First note that majority is odd when n is odd:

$$\mathrm{Maj}_n(-u) = -\mathrm{Maj}_n(u),$$

because flipping all signs flips the strict majority.

- Base case $d = 1$: true for Maj_n.
- Inductive step: assuming $RM_{d-1}(-z) = -RM_{d-1}(z)$,

$$RM_d(-x) = \mathrm{Maj}_n\left(RM_{d-1}(-x^{(1)}), \ldots, RM_{d-1}(-x^{(n)})\right)$$

$$= \mathrm{Maj}_n\left(-RM_{d-1}(x^{(1)}), \ldots, -RM_{d-1}(x^{(n)})\right) = -RM_d(x).$$

So RM_d is odd.

3. Unanimous: Yes

If all inputs are $+1$, then every depth-$(d-1)$ block evaluates to $+1$, so the top majority is $+1$. Likewise, if all inputs are -1, then every block evaluates to -1, so the top majority is -1.

Formally, by induction:

- $RM_1(1,\ldots,1) = 1$, $RM_1(-1,\ldots,-1) = -1$.
- If true for $d-1$, then

$$RM_d(1,\ldots,1) = \mathrm{Maj}_n(1,\ldots,1) = 1,$$

and similarly

$$RM_d(-1,\ldots,-1) = \mathrm{Maj}_n(-1,\ldots,-1) = -1.$$

So RM_d is unanimous.

4. Symmetric: Only for $d = 1$; not for $d \geq 2$

A symmetric Boolean function depends only on the number of $+1$'s, not on where they occur.

- For $d = 1$, $RM_1 = \mathrm{Maj}_n$, which is symmetric.
- For $d \geq 2$, recursive majority is **not** symmetric, because the block structure matters.

Counterexample for $d = 2$

Case $n = 3$
Take two inputs on 9 bits with the same number of $+1$'s.

- x: the three top-level blocks contain $2, 2, 0$ many $+1$'s.
  Then the block outputs are $+, +, -$, so $RM_2(x) = +1$.
- y: the three top-level blocks contain $3, 1, 0$ many $+1$'s.
  Then the block outputs are $+, -, -$, so $RM_2(y) = -1$.

Both have 4 many $+1$'s, so one is a permutation of the other, yet the outputs differ. Hence RM_2 is not symmetric.

Case $n \geq 5$
Let $m = \frac{n+1}{2}$.

Construct two inputs with the same total number of $+1$'s:

- x: exactly m top-level blocks are all $+1$, and the remaining $n - m$ blocks are all -1.
  Then m block outputs are $+1$, so $RM_2(x) = +1$.
  Total number of $+1$'s is mn.
- y: $m - 1$ top-level blocks are all $+1$; among the remaining m blocks, two blocks contain exactly $m - 1$ many $+1$'s, one block contains exactly one $+1$, and the rest are all -1.
  Then the total number of $+1$'s is

$$(m-1)n + 2(m-1) + 1 = mn,$$

the same as for x.
But only $m-1$ blocks have positive majority, so $RM_2(y) = -1$.

Thus x and y have the same Hamming weight but different outputs, so RM_2 is not symmetric.

For $d > 2$

Embed the $d = 2$ counterexample by replacing each of its coordinates with a constant block of size n^{d-2}. Each such constant block evaluates under RM_{d-2} to its constant sign, so the depth-d recursive majority on the expanded input behaves exactly like the depth-2 example. Hence RM_d is also not symmetric for every $d \geq 2$.

Final answer

For recursive majority of depth d:

1. **Monotone:** Yes
2. **Odd:** Yes
3. **Unanimous:** Yes
4. **Symmetric: Yes if $d = 1$, no if $d \geq 2$**

Problem 4 (Influence of the monotone chain function). Let $f : \{-1,1\}^n \to \{-1,1\}$ be defined by

$$f(x) = 1 \quad \text{if and only if} \quad x_1 \leq x_2 \leq \cdots \leq x_n.$$

Compute the total influence $I[f]$ exactly as a function of n.

Thought for 1m 58s ›

Let

$$A := \{x \in \{-1,1\}^n : \; x_1 \leq x_2 \leq \cdots \leq x_n\}.$$

Then $f(x) = 1$ exactly on A, and $f(x) = -1$ off A.

For $\{-1,1\}$-valued functions, the total influence is

$$I[f] = \sum_{i=1}^{n} \Pr_x \left[f(x) \neq f(x^{\oplus i}) \right],$$

where $x^{\oplus i}$ is x with the i-th bit flipped.

Step 1: Describe the 1-set of f

Because coordinates are only ± 1, the condition $x_1 \leq x_2 \leq \cdots \leq x_n$ means x must look like

$$(-1, \dots, -1, 1, \dots, 1).$$

So there are exactly $n+1$ such points:

$$a_k := (\underbrace{-1, \dots, -1}_{k}, \underbrace{1, \dots, 1}_{n-k}), \qquad k = 0, 1, \dots, n.$$

Hence $|A| = n + 1$.

Also, a_k and a_{k-1} differ in exactly one coordinate, so the induced subgraph on A is a path of length n. Thus

$$|E(A)| = n.$$

Step 2: Count bichromatic edges

A bichromatic edge is a hypercube edge joining a point of A to a point outside A. These are exactly the edges contributing to influence.

Each vertex in the n-cube has degree n, so the total number of edge-incidences from vertices of A is

$$n|A| = n(n+1).$$

Among these, each internal edge of A is counted twice, and there are n such internal edges. Therefore the number B of bichromatic edges is

$$B = n|A| - 2|E(A)| = n(n+1) - 2n = n(n-1).$$

Step 3: Convert edge count to total influence

In

$$\sum_{x \in \{-1,1\}^n} \sum_{i=1}^{n} \mathbf{1}\{f(x) \neq f(x^{\oplus i})\},$$

each bichromatic edge is counted twice, once from each endpoint. Hence

$$I[f] = \frac{1}{2^n} \cdot 2B = \frac{B}{2^{n-1}} = \frac{n(n-1)}{2^{n-1}}.$$

So the exact total influence is

$$\boxed{I[f] = \frac{n(n-1)}{2^{n-1}}}.$$

As a bonus, for each fixed coordinate i, there are exactly $n-1$ bichromatic edges in direction i, so

$$\backslash\mathrm{Inf}_i[f] = \frac{n-1}{2^{n-1}}, \qquad i = 1, \dots, n,$$

and summing over i again gives

$$I[f] = n \cdot \frac{n-1}{2^{n-1}} = \frac{n(n-1)}{2^{n-1}}.$$

Problem 1 (Concentration of Fourier spectrum). Suppose the Fourier spectrum of $f : \{-1,1\}^n \to \mathbb{R}$ is ϵ-concentrated on $\mathcal{F}$. Let $g : \{-1,1\}^n \to \mathbb{R}$ be a function satisfying $\|f - g\|_2^2 \leq \epsilon$. Prove that the Fourier spectrum of g is $O(\epsilon)$-concentrated on $\mathcal{F}$.

Thought for 15s ›

Let

$$f(x) = \sum_{S \subseteq [n]} \hat{f}(S) \chi_S(x), \qquad g(x) = \sum_{S \subseteq [n]} \hat{g}(S) \chi_S(x),$$

and assume that "the Fourier spectrum of f is ϵ-concentrated on $\mathcal{F}$" means

$$\sum_{S \notin \mathcal{F}} \hat{f}(S)^2 \leq \epsilon.$$

Also, by Parseval,

$$\|f - g\|_2^2 = \sum_{S \subseteq [n]} (\hat{f}(S) - \hat{g}(S))^2 \leq \epsilon.$$

We want to bound the Fourier mass of g outside $\mathcal{F}$.

For any $S \notin \mathcal{F}$,

$$\hat{g}(S) = \hat{f}(S) + (\hat{g}(S) - \hat{f}(S)).$$

Hence, using $(a+b)^2 \leq 2a^2 + 2b^2$,

$$\hat{g}(S)^2 \leq 2\hat{f}(S)^2 + 2(\hat{g}(S) - \hat{f}(S))^2.$$

Summing over $S \notin \mathcal{F}$,

$$\sum_{S \notin \mathcal{F}} \hat{g}(S)^2 \leq 2 \sum_{S \notin \mathcal{F}} \hat{f}(S)^2 + 2 \sum_{S \notin \mathcal{F}} (\hat{g}(S) - \hat{f}(S))^2.$$

Now use the assumptions:

$$\sum_{S \notin \mathcal{F}} \hat{f}(S)^2 \leq \epsilon,$$

and

$$\sum_{S \notin \mathcal{F}} (\hat{g}(S) - \hat{f}(S))^2 \leq \sum_{S \subseteq [n]} (\hat{g}(S) - \hat{f}(S))^2 = \|f - g\|_2^2 \leq \epsilon.$$

Therefore,

$$\sum_{S \notin \mathcal{F}} \hat{g}(S)^2 \leq 2\epsilon + 2\epsilon = 4\epsilon.$$

So the Fourier spectrum of g is 4ϵ-concentrated on $\mathcal{F}$, and therefore $O(\epsilon)$-concentrated on $\mathcal{F}$.

$$\boxed{\sum_{S \notin \mathcal{F}} \hat{g}(S)^2 \leq 4\epsilon.}$$

So g is $O(\epsilon)$-concentrated on $\mathcal{F}$.

Problem 2 (Pseudorandom generator via one-way permutation). Let $f : \mathbb{F}_2^n \to \mathbb{F}_2^n$ be a permutation. Suppose there exists a efficient deterministic algorithm A and a parameter $\gamma > 0$ such that

$$\Pr_{\boldsymbol{r}, \boldsymbol{s} \sim \mathbb{F}_2^n} [A(\boldsymbol{r}, f(\boldsymbol{s})) = \boldsymbol{r} \cdot \boldsymbol{s}] \geq \frac{1}{2} + \gamma,$$

where $\boldsymbol{r} \cdot \boldsymbol{s} = \sum_{i=1}^n \boldsymbol{r}_i \boldsymbol{s}_i \pmod 2$ denotes the inner product over $\mathbb{F}_2$.

Show that one can use A together with the Goldreich–Levin algorithm to recover s from $f(s)$ for a $\gamma/2$ fraction of $s \in \mathbb{F}_2^n$ with non-negligible probability.

More precisely, prove the following steps.

1. Show that for at least a $\gamma/2$ fraction of strings $s \in \mathbb{F}_2^n$,

$$\Pr_{\boldsymbol{r} \sim \mathbb{F}_2^n} [A(\boldsymbol{r}, f(s)) = \boldsymbol{r} \cdot s] \geq \frac{1}{2} + \frac{\gamma}{2}.$$

2. Fix any such s in the subsequent discussion. Define

$$h_s(r) := (-1)^{A(r, f(s))}, \qquad r \in \mathbb{F}_2^n.$$

Let

$$S := \{ i \in [n] : s_i = 1 \}.$$

Viewing h_s as a Boolean function on $\{-1,1\}^n$, prove that

$$\widehat{h_s}(S) \geq \gamma.$$

3. Explain how to use the Goldreich–Levin algorithm to efficiently recover s from $f(s)$ with high probability.

Thought for 1m 31s ›

Let

$$p_s := \Pr_{r\in\mathbb{F}_2^n}\left[A(r,f(s)) = r\cdot s\right].$$

The hypothesis says

$$\mathbb{E}_s[p_s] = \Pr_{r,s}[A(r,f(s)) = r\cdot s] \ge \frac{1}{2} + \gamma.$$

We prove the three requested steps.

---

1. A $\gamma/2$ fraction of s's are "good"

Call s **good** if

$$p_s \ge \frac{1}{2} + \frac{\gamma}{2}.$$

We claim that at least a $\gamma/2$ fraction of $s \in \mathbb{F}_2^n$ are good.

Assume not. Then fewer than a $\gamma/2$ fraction are good, so more than a $1-\gamma/2$ fraction satisfy

$$p_s < \frac{1}{2} + \frac{\gamma}{2}.$$

Since always $p_s \le 1$, we get

$$\mathbb{E}_s[p_s] < \frac{\gamma}{2}\cdot 1 + \left(1-\frac{\gamma}{2}\right)\left(\frac{1}{2}+\frac{\gamma}{2}\right).$$

Expanding,

$$\mathbb{E}_s[p_s] < \frac{\gamma}{2} + \frac{1}{2} + \frac{\gamma}{2} - \frac{\gamma}{4} - \frac{\gamma^2}{4} = \frac{1}{2} + \frac{3\gamma}{4} - \frac{\gamma^2}{4} < \frac{1}{2} + \gamma,$$

contradicting the hypothesis.

Therefore,

$$\Pr_s\left[p_s \ge \frac{1}{2} + \frac{\gamma}{2}\right] \ge \frac{\gamma}{2}.$$

So at least a $\gamma/2$ fraction of strings s satisfy

$$\Pr_r[A(r,f(s)) = r\cdot s] \ge \frac{1}{2} + \frac{\gamma}{2}.$$

Define the oracle function

$$h_y(r) := (-1)^{A(r,y)}.$$

This is efficiently computable from y, since A is efficient.
If $y = f(s)$ for a good s, then by Step 2 the function $h_y = h_s$ has a Fourier coefficient at the set

$$S = \{i : s_i = 1\}$$

of magnitude at least γ:

$$\hat{h}_y(S) = \hat{h}_s(S) \ge \gamma.$$

Now run the Goldreich–Levin algorithm on h_y with threshold, say,

$$\theta = \frac{\gamma}{2}.$$

Goldreich–Levin outputs, in time $\mathrm{poly}(n, 1/\gamma, \log(1/\delta))$, a list L of subsets $T \subseteq [n]$ such that, with probability at least $1-\delta$, every T with $|\hat{h}_y(T)| \ge \theta$ appears in the list. Since $\hat{h}_y(S) \ge \gamma > \gamma/2$, the true set S is in L with probability at least $1-\delta$.

From each $T \in L$, form the candidate string $s_T \in \mathbb{F}_2^n$ defined by

$$(s_T)_i = \begin{cases} 1 & i \in T, \\ 0 & i \notin T. \end{cases}$$

Then check which candidate satisfies

$$f(s_T) = y.$$

Because f is a permutation, there is exactly one preimage of y, namely the true s. So once $S \in L$, this verification step identifies s uniquely.

So the recovery algorithm is:

1. Input y.
2. Define oracle $h_y(r) = (-1)^{A(r,y)}$.
3. Run Goldreich–Levin on h_y with threshold $\gamma/2$.
4. For each returned subset T, test whether $f(s_T) = y$.
5. Output the unique s_T that passes.

This is efficient, and for every good s, it recovers s with probability at least $1-\delta$. Since at least a $\gamma/2$ fraction of s's are good, we conclude:

$$\Pr_s\left[\text{the algorithm recovers } s \text{ from } f(s)\right] \ge \frac{\gamma}{2}(1-\delta).$$

2. For such an s, the corresponding Fourier coefficient is large

Fix a good s, and let

$$h_s(r) := (-1)^{A(r,f(s))}, \qquad r \in \mathbb{F}_2^n.$$

Let

$$S := \{i \in [n] : s_i = 1\}.$$

We now view h_s as a Boolean function on $\{-1,1\}^n$.
Identify $r \in \mathbb{F}_2^n$ with $x(r) \in \{-1,1\}^n$ by

$$x_i(r) = (-1)^{r_i}.$$

Under this identification, the character χ_S is

$$\chi_S(x(r)) = \prod_{i\in S} x_i(r) = (-1)^{\sum_{i\in S} r_i} = (-1)^{r\cdot s}.$$

Hence the Fourier coefficient of h_s at S is

$$\hat{h}_s(S) = \mathbb{E}_r\left[h_s(r)\chi_S(x(r))\right] = \mathbb{E}_r\left[(-1)^{A(r,f(s))}(-1)^{r\cdot s}\right].$$

Combine the exponents:

$$\hat{h}_s(S) = \mathbb{E}_r\left[(-1)^{A(r,f(s))+r\cdot s}\right].$$

The random variable $(-1)^{A(r,f(s))+r\cdot s}$ equals $+1$ when $A(r,f(s)) = r\cdot s$, and -1 otherwise. Therefore

$$\hat{h}_s(S) = \Pr_r[A(r,f(s)) = r\cdot s] - \Pr_r[A(r,f(s)) \ne r\cdot s] = 2p_s - 1.$$

Since s is good, $p_s \ge \frac{1}{2} + \frac{\gamma}{2}$, so

$$\hat{h}_s(S) \ge 2\left(\frac{1}{2}+\frac{\gamma}{2}\right) - 1 = \gamma.$$

Thus

$$\boxed{\hat{h}_s(S) \ge \gamma.}$$

---

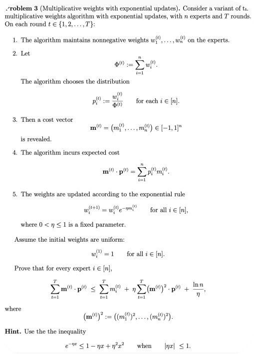
3. Use Goldreich–Levin to recover s from $f(s)$

Now suppose we are given

$$y = f(s).$$

Thus one can recover s from $f(s)$ for at least a $\gamma/2$ fraction of inputs with non-negligible probability.

If you'd like, I can also turn this into a shorter "homework-style" writeup with fewer words and just the essential equations.


Problem 3 (Multiplicative weights with exponential updates). Consider a variant of the multiplicative weights algorithm with exponential updates, with n experts and T rounds. On each round $t \in \{1,2,\ldots,T\}$:

1. The algorithm maintains nonnegative weights $w_1^{(t)},\ldots,w_n^{(t)}$ on the experts.
2. Let
$$\Phi^{(t)} := \sum_{i=1}^n w_i^{(t)}.$$
The algorithm chooses the distribution
$$p_i^{(t)} := \frac{w_i^{(t)}}{\Phi^{(t)}} \qquad \text{for each } i \in [n].$$
3. Then a cost vector
$$\mathbf{m}^{(t)} = \left(m_1^{(t)},\ldots,m_n^{(t)}\right) \in [-1,1]^n$$
is revealed.
4. The algorithm incurs expected cost
$$\mathbf{m}^{(t)}\cdot\mathbf{p}^{(t)} = \sum_{i=1}^n p_i^{(t)} m_i^{(t)}.$$
5. The weights are updated according to the exponential rule
$$w_i^{(t+1)} = w_i^{(t)} e^{-\eta m_i^{(t)}} \qquad \text{for all } i \in [n],$$
where $0 < \eta \le 1$ is a fixed parameter.

Assume the initial weights are uniform:

$$w_i^{(1)} = 1 \qquad \text{for all } i \in [n].$$

Prove that for every expert $i \in [n]$,

$$\sum_{t=1}^T \mathbf{m}^{(t)}\cdot\mathbf{p}^{(t)} \le \sum_{t=1}^T m_i^{(t)} + \eta\sum_{t=1}^T (\mathbf{m}^{(t)})^2\cdot\mathbf{p}^{(t)} + \frac{\ln n}{\eta},$$

where

$$(\mathbf{m}^{(t)})^2 := \left((m_1^{(t)})^2,\ldots,(m_n^{(t)})^2\right).$$

Hint. Use the the inequality

$$e^{-\eta x} \le 1 - \eta x + \eta^2 x^2 \qquad \text{when} \qquad |\eta x| \le 1.$$

Thought for 25s ›

Let

$$L_A := \sum_{t=1}^T \mathbf{m}^{(t)}\cdot\mathbf{p}^{(t)} \qquad \text{and} \qquad Q := \sum_{t=1}^T (\mathbf{m}^{(t)})^2\cdot\mathbf{p}^{(t)}.$$

We will prove that for every expert i,

$$L_A \le \sum_{t=1}^T m_i^{(t)} + \eta Q + \frac{\ln n}{\eta}.$$

The proof is by the usual potential argument.

---

1. Potential evolution

Recall

$$\Phi^{(t)} = \sum_{j=1}^{n} w_j^{(t)}, \qquad p_j^{(t)} = \frac{w_j^{(t)}}{\Phi^{(t)}}, \qquad w_j^{(t+1)} = w_j^{(t)} e^{-\eta m_j^{(t)}}.$$

Then

$$\Phi^{(t+1)} = \sum_{j=1}^{n} w_j^{(t+1)} = \sum_{j=1}^{n} w_j^{(t)} e^{-\eta m_j^{(t)}}.$$

Since $m_j^{(t)} \in [-1, 1]$ and $0 < \eta \le 1$, we have $|\eta m_j^{(t)}| \le 1$, so the hint gives

$$e^{-\eta m_j^{(t)}} \le 1 - \eta m_j^{(t)} + \eta^2 (m_j^{(t)})^2.$$

Therefore,

$$\Phi^{(t+1)} \le \sum_{j=1}^{n} w_j^{(t)} \left(1 - \eta m_j^{(t)} + \eta^2 (m_j^{(t)})^2\right).$$

Pulling out $\Phi^{(t)}$,

$$\Phi^{(t+1)} \le \Phi^{(t)} \left(1 - \eta \sum_{j=1}^{n} p_j^{(t)} m_j^{(t)} + \eta^2 \sum_{j=1}^{n} p_j^{(t)} (m_j^{(t)})^2\right).$$

That is,

$$\Phi^{(t+1)} \le \Phi^{(t)} \left(1 - \eta\, \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} + \eta^2\, (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)}\right).$$

Using $1 + z \le e^z$,

$$\Phi^{(t+1)} \le \Phi^{(t)} \exp\left(-\eta\, \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} + \eta^2\, (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)}\right).$$

Taking logs,

$$\ln \Phi^{(t+1)} - \ln \Phi^{(t)} \le -\eta\, \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} + \eta^2\, (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)}.$$

Summing from $t = 1$ to T,

$$\ln \Phi^{(T+1)} - \ln \Phi^{(1)} \le -\eta \sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} + \eta^2 \sum_{t=1}^{T} (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)}.$$

Since $w_j^{(1)} = 1$, we have $\Phi^{(1)} = n$. Hence

$$\ln \Phi^{(T+1)} - \ln n \le -\eta L_A + \eta^2 Q. \tag{1}$$

---

2. Lower bound the final potential using one expert

Fix any expert i. Its weight evolves as

$$w_i^{(T+1)} = w_i^{(1)} \prod_{t=1}^{T} e^{-\eta m_i^{(t)}} = \exp\left(-\eta \sum_{t=1}^{T} m_i^{(t)}\right),$$

because $w_i^{(1)} = 1$.

Since $\Phi^{(T+1)} = \sum_j w_j^{(T+1)} \ge w_i^{(T+1)}$,

$$\ln \Phi^{(T+1)} \ge -\eta \sum_{t=1}^{T} m_i^{(t)}. \tag{2}$$

---

3. Combine

Plug (2) into (1):

$$-\eta \sum_{t=1}^{T} m_i^{(t)} - \ln n \le -\eta L_A + \eta^2 Q.$$

Rearrange:

$$\eta L_A \le \eta \sum_{t=1}^{T} m_i^{(t)} + \eta^2 Q + \ln n.$$

Divide by $\eta > 0$:

$$L_A \le \sum_{t=1}^{T} m_i^{(t)} + \eta \sum_{t=1}^{T} (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)} + \frac{\ln n}{\eta}.$$

Since $L_A = \sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)}$, this is exactly

$$\boxed{\sum_{t=1}^{T} \mathbf{m}^{(t)} \cdot \mathbf{p}^{(t)} \le \sum_{t=1}^{T} m_i^{(t)} + \eta \sum_{t=1}^{T} (\mathbf{m}^{(t)})^2 \cdot \mathbf{p}^{(t)} + \frac{\ln n}{\eta}}.$$

This holds for every expert $i \in [n]$.